

Basic SQL SELECT: Project Report

Dataset: Categories, Customers, Products, Suppliers, Shippers, Orders, OrderDetails | Tools: MySQL / PostgreSQL syntax | 28 September 2026

1. Objective

Build SQL fundamentals by writing 10 simple queries using SELECT, WHERE and ORDER BY on a Northwind-style database. Before the queries, the seven CSV files were cleaned and checked. All queries below were run on the cleaned data, and the outputs shown are the actual results.

2. Data cleaning summary

File	Rows	Issue found	Fix applied
Categories	21	Unquoted commas in Description; trailing spaces	Rebuilt columns, trimmed, quoted text
Customers	23	Rows 5 and 17 had extra columns; trailing spaces	Rejoined address (ID 5); removed stray 'JA' (ID 17); trimmed
Products	25	Trailing spaces made Price text	Trimmed; Price numeric, 2 decimals
Suppliers	22	Trailing spaces in Phone	Trimmed
Shippers	22	Trailing spaces in Phone	Trimmed
Orders	21	Trailing spaces in ShipperID	Trimmed; IDs to integers; dates checked (all valid)
OrderDetails	26	Trailing spaces in Quantity	Trimmed; IDs and Quantity to integers

3. Data integrity checks

Check	Result
Primary keys unique in all 7 files	Passed
Same column count on every row, all files	Passed
Products.SupplierID and CategoryID exist in Suppliers and Categories	Passed
Orders.ShipperID exists in Shippers	Passed
OrderDetails.OrderID exists in Orders	Passed
Orders.CustomerID exists in Customers	18 of 21 orders do not match (Orders use CustomerID up to 99; Customers has 23 rows)
OrderDetails.ProductID exists in Products	17 of 26 lines do not match (lines use ProductID up to 72; Products has 25 rows)

What this means: the 10 queries below each read from a single table, so they are not affected. But a JOIN between Orders and Customers, or OrderDetails and Products, will drop most rows. To fix that, the Customers and Products files would need the missing IDs added.

4. Queries and outputs

Query 1: List all product categories

```
SELECT * FROM Categories;
```

CategoryID	CategoryName	Description
1	Beverages	Soft drinks, coffees, teas, beers, and ales

CategoryID	CategoryName	Description
2	Condiments	Sweet and savory sauces, relishes, spreads, and seasonings
3	Confections	Desserts, candies, and sweet breads
4	Dairy Products	Cheeses
5	Grains/Cereals	Breads, crackers, pasta, and cereal
6	Seafood	Fish and shellfish products, both fresh and preserved
7	Fruits and Vegetables	Fresh and preserved fruits and vegetables
8	Nuts and Seeds	Various types of nuts and seeds, both raw and roasted
9	Meat	Pork, beef, poultry, and processed meats
10	Bakery	Freshly baked breads and pastries
11	Snacks	Chips, popcorn, and other savory snack foods
12	Canned Goods	Preserved fruits, vegetables, and meats in jars or cans
13	Spices & Seasonings	Dry spices, herbs, and blends for cooking
14	Healthy Foods	Organic, gluten-free, and health-oriented food options
15	Prepared Meals	Cooked and ready-to-eat meals and entrees
16	Ice Cream & Frozen Desserts	Frozen sweets, including ice cream and sorbet
17	Ethnic Foods	Specialty foods from various world cuisines
18	Desserts	Baked goods like pies, cakes, and pastries
19	Coffees & Teas	Selection of gourmet and artisanal coffee and tea
20	Organic Foods	Certified organic products across multiple categories
21	Breakfast Foods	Cereals, oatmeal, and breakfast beverages

21 row(s) returned.

Query 2: Customers from Germany

```
SELECT CustomerName, City, Country
FROM Customers
WHERE Country = 'Germany';
```

CustomerName	City	Country
Alfreds Futterkiste	Berlin	Germany
Blauer See Delikatessen	Mannheim	Germany
Feinkost Käfer	München	Germany
Gourmet Delights	Berlin	Germany

4 row(s) returned.

Query 3: Products priced above 20

```
SELECT ProductName, Price
FROM Products
WHERE Price > 20;
```

ProductName	Price
Chef Anton's Cajun Seasoning	22.00
Chef Anton's Gumbo Mix	21.35
Grilled Chicken	30.00
Jasmine Rice	25.00
Clam Chowder	22.00
Maple Syrup	25.00
Vanilla Extract	30.00

7 row(s) returned.

Query 4: Products from most to least expensive

```
SELECT ProductName, Price
FROM Products
ORDER BY Price DESC;
```

ProductName	Price
Grilled Chicken	30.00
Vanilla Extract	30.00
Jasmine Rice	25.00
Maple Syrup	25.00
Chef Anton's Cajun Seasoning	22.00
Clam Chowder	22.00
Chef Anton's Gumbo Mix	21.35
Vegetable Stir-Fry Mix	19.99
Chang	19.00
Chais	18.00
Chocolate Sauce	18.00
Granola Bars	16.00
Tomato Sauce	15.00
Peanut Butter	15.00
Apple Juice	14.00
Green Tea	12.00
Almond Flour	11.50
Sesame Oil	11.00
Aniseed Syrup	10.00
Mango Chutney	9.95
Instant Coffee	9.50
Basil Pesto	7.20
Organic Oats	4.50
Whole Wheat Pasta	3.50

ProductName	Price
Cream Cheese	2.70

25 row(s) returned.

Query 5: US suppliers sorted by city

```
SELECT SupplierName, City, Country
FROM Suppliers
WHERE Country = 'USA'
ORDER BY City ASC;
```

SupplierName	City	Country
Grandma Kelly's Homestead	Ann Arbor	USA
Ice Cream Dream	Chicago	USA
Natural Nut Company	Los Angeles	USA
Sweet Honey Farms	Memphis	USA
New Orleans Cajun Delights	New Orleans	USA
Kingfish Seafood	New Orleans	USA
Peanut Butter Paradise	Orlando	USA
Yummy Cupcakes	San Diego	USA
Fruitful Farms	San Francisco	USA

9 row(s) returned.

Query 6: Condiments (CategoryID 2) priced under 20

```
SELECT ProductName, CategoryID, Price
FROM Products
WHERE CategoryID = 2 AND Price < 20
ORDER BY Price;
```

ProductName	CategoryID	Price
Basil Pesto	2	7.20
Mango Chutney	2	9.95
Aniseed Syrup	2	10.00
Sesame Oil	2	11.00
Tomato Sauce	2	15.00
Peanut Butter	2	15.00
Chocolate Sauce	2	18.00

7 row(s) returned.

Query 7: Customers whose name starts with 'A'

```
SELECT CustomerName, Country
FROM Customers
WHERE CustomerName LIKE 'A%'
ORDER BY CustomerName;
```

CustomerName	Country
Alfreds Futterkiste	Germany
Ana Trujillo Emparedados y helados	Mexico
Antonio Moreno Taquería	Mexico

3 row(s) returned.

Query 8: First five shippers, sorted by name

```
SELECT ShipperID, CompanyName, Phone
FROM Shippers
WHERE ShipperID <= 5
ORDER BY CompanyName;
```

ShipperID	CompanyName	Phone
5	Express Delivery	(503) 555-8765
4	Fast Freight	(503) 555-2244
3	Federal Shipping	(503) 555-9931
1	Speedy Express	(503) 555-9831
2	United Package	(503) 555-3199

5 row(s) returned.

Query 9: Orders placed between 10 and 20 July 2022

```
SELECT OrderID, CustomerID, OrderDate
FROM Orders
WHERE OrderDate BETWEEN '2022-07-10' AND '2022-07-20'
ORDER BY OrderDate;
```

OrderID	CustomerID	OrderDate
10253	50	2022-07-10
10254	89	2022-07-11
10255	12	2022-07-12
10256	73	2022-07-13
10257	22	2022-07-14
10258	11	2022-07-15
10259	45	2022-07-16
10260	78	2022-07-17
10261	39	2022-07-18
10262	41	2022-07-19
10263	54	2022-07-20

11 row(s) returned.

Query 10: Five largest order lines by quantity

```
SELECT OrderID, ProductID, Quantity
FROM OrderDetails
ORDER BY Quantity DESC
LIMIT 5;
```

OrderID	ProductID	Quantity
10249	51	40
10254	29	22
10250	31	20
10256	12	18
10261	13	16

5 row(s) returned.

5. Interview questions

What does SELECT do? It retrieves data from one or more tables, either chosen columns or every column with *.

What does ORDER BY do? It sorts the result set by one or more columns, ascending (ASC, the default) or descending (DESC).

What does WHERE do? It filters rows so only those meeting a condition are returned.